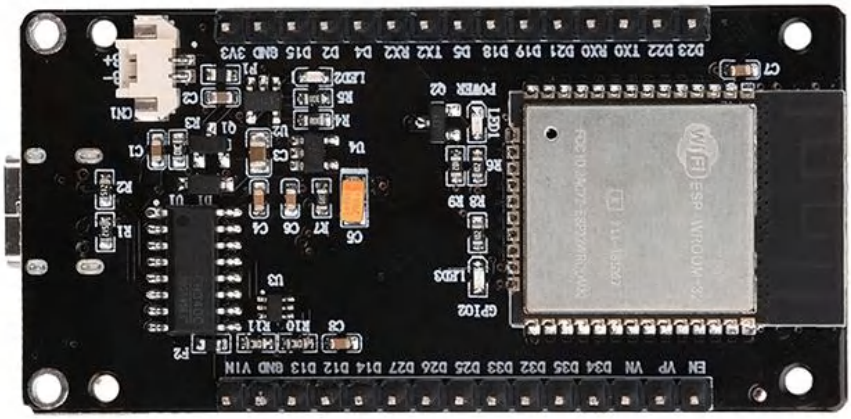
ESP32 1.9″ LCD screen Development board



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Product Introduction

This development board integrates ESP32 and a 1.9-inch LCD screen, providing all the functions of the traditional ESP32 module, the same peripheral ports, as well as high-performance development tools and high-definition data display. It supports multiple programming modes, and the 2.4GHz WiFi Bluetooth function enables it to easily handle various tasks with a complete network protocol stack and security mechanism. It has low power consumption and high performance, suitable for beginners and professionals to develop and apply loT projects, plug and play. It supports multiple interfaces such as UART,SPI, SDIO, I2C, PWM, I2S,lR, ADC, and DAC for easy external connection. It has a wide range of application scenarios and is suitable for WiFi solutions, WiFi Mini cameras, Mesh networking, smart homes, mobile loT, etc.

The ESP32-WROOM-32 module on this development board is a universal Wi-Fi+BT+BLE MCU module with low-power Bluetooth and Wi-Fi. lt has a wide range of uses and integrates SPI flash memory. Wi-Fi supports a wide range of wireless communication connections andsupports direct connection to the Internet through a router; while Bluetooth allows users toconnect to mobile phones for easy signal detection. Application scenarios: WiFi solutions, WiFi Mini cameras, Mesh networking, smart homes, mobile Internet of Things, etc., plug and play.

Product parameters:

lnterface: TYPE-C

Working voltage: USB 5V

Driver chip: CH340

LCD display: 1.9 inches

SPI Flash: default 32Mbit

Serial port rate: 115200bps

Frequency range: 2412-2484MHz

Bluetooth protocol: Bluetooth 4.2 BR/EDR and BLE standard

WiFi protocol: 802.11 b/g/n

Antenna form: onboard PCB antenna, gain 2dBi

Supported interfaces: UART,SPI,SDIO, I2C, PWM, I2S, IR, ADC,DAC

EN: Enable reset button

BOOT: Download button Press the Boot button and hold (do not release the Boot button at this time) to enter the "firmware download" mode and download the firmware through the serial port

USB: can be used as a power supply for the circuit board,or connect the PC and ESP32-WROOM-32 Communication interface of the module

Size: 63.70mm*31.5mm

Product list:

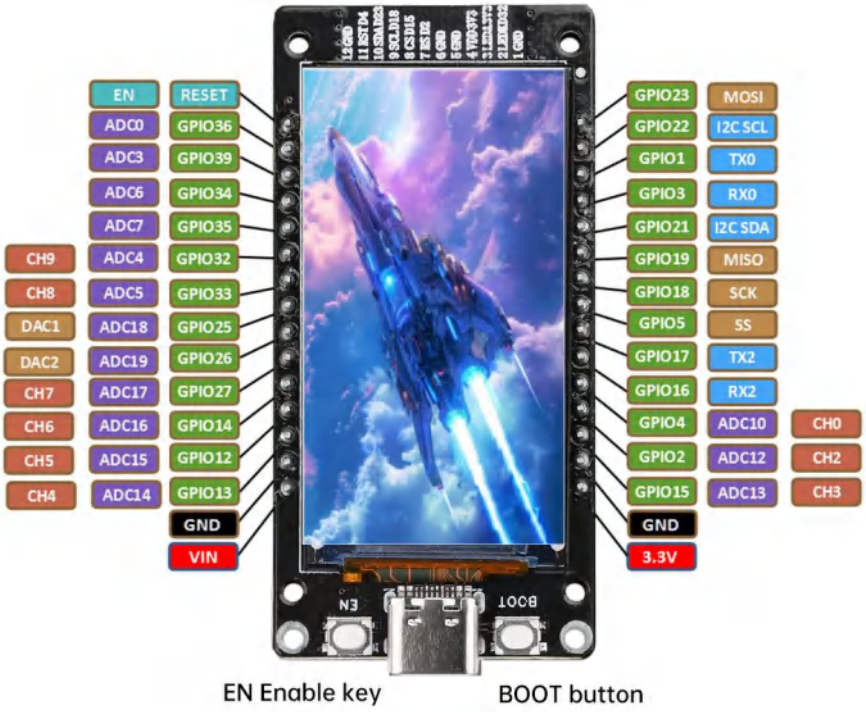
1x development board

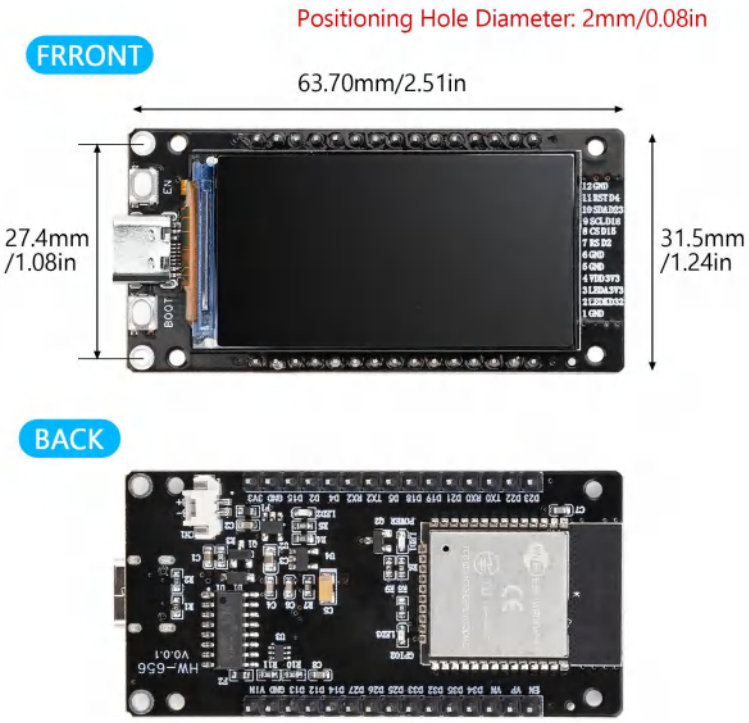
Note:

Please connect the wires according to the diagram and do not make mistakes.

This document only represents the product knowledge and parameters at the time of editing. lf there are any changes in the future, we will not notify you separately.

Product Pin Description





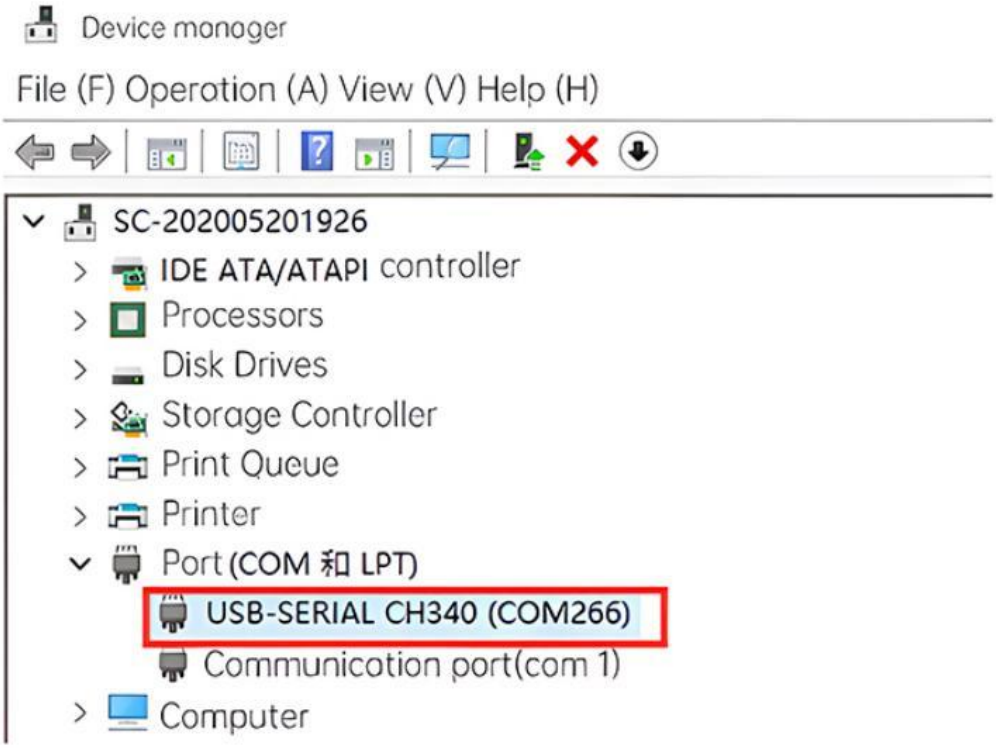
Instructions

lnstructions for use (taking LCD display as an example):

Connect the development board to the computer via a Type-C data cable (the power indicator light turns on after power on)

Open My Computer-Management-Device Manager to check if the COM port recognition is

normal



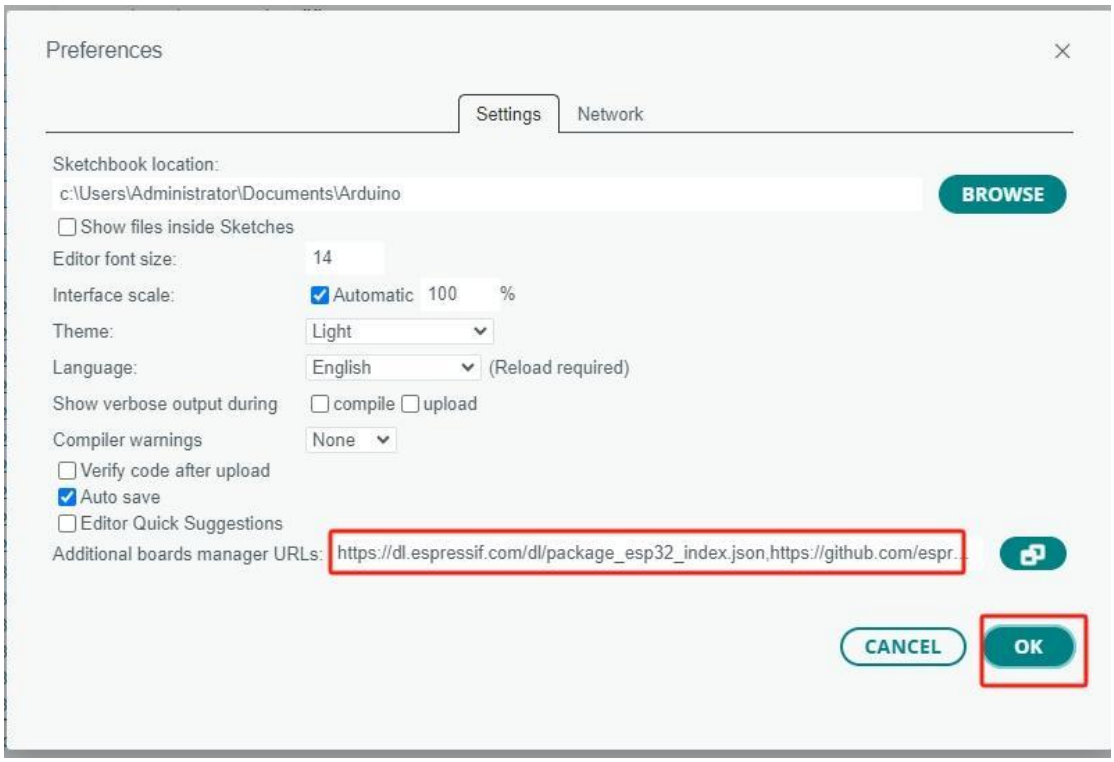
LCD demo download

1. Open Arduino IDE 1.8.5, find File →Preferences, then copy and paste the following link into the Preferences Additional URL, as shown in the red box, and click "OK"

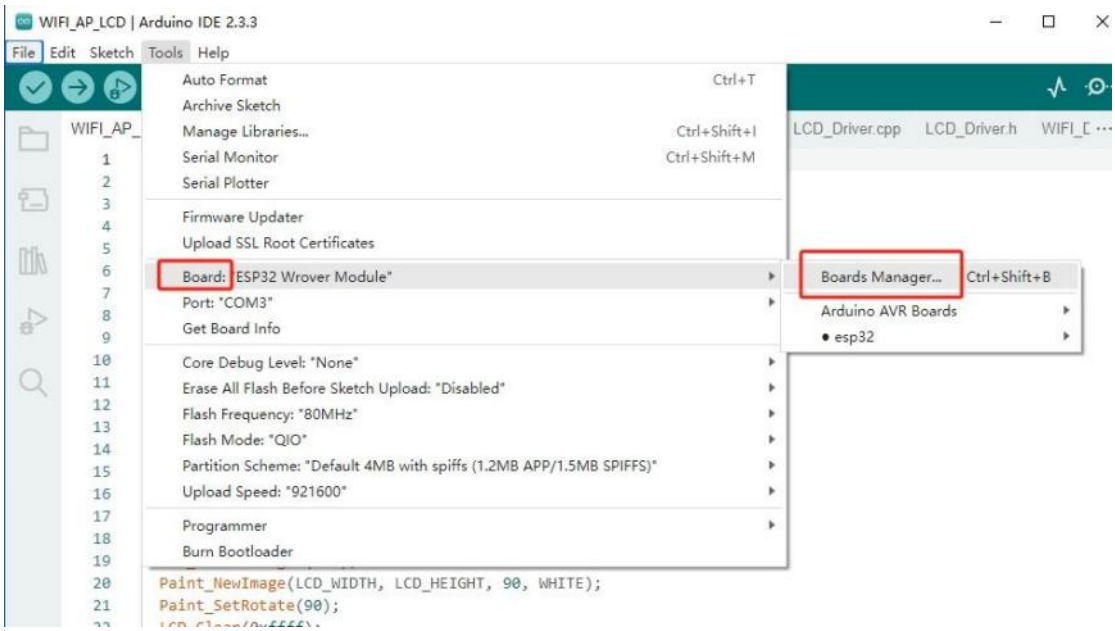
https://dl.espressif.com/dl/package_esp32_index.json

For more product library information, open this link to download.

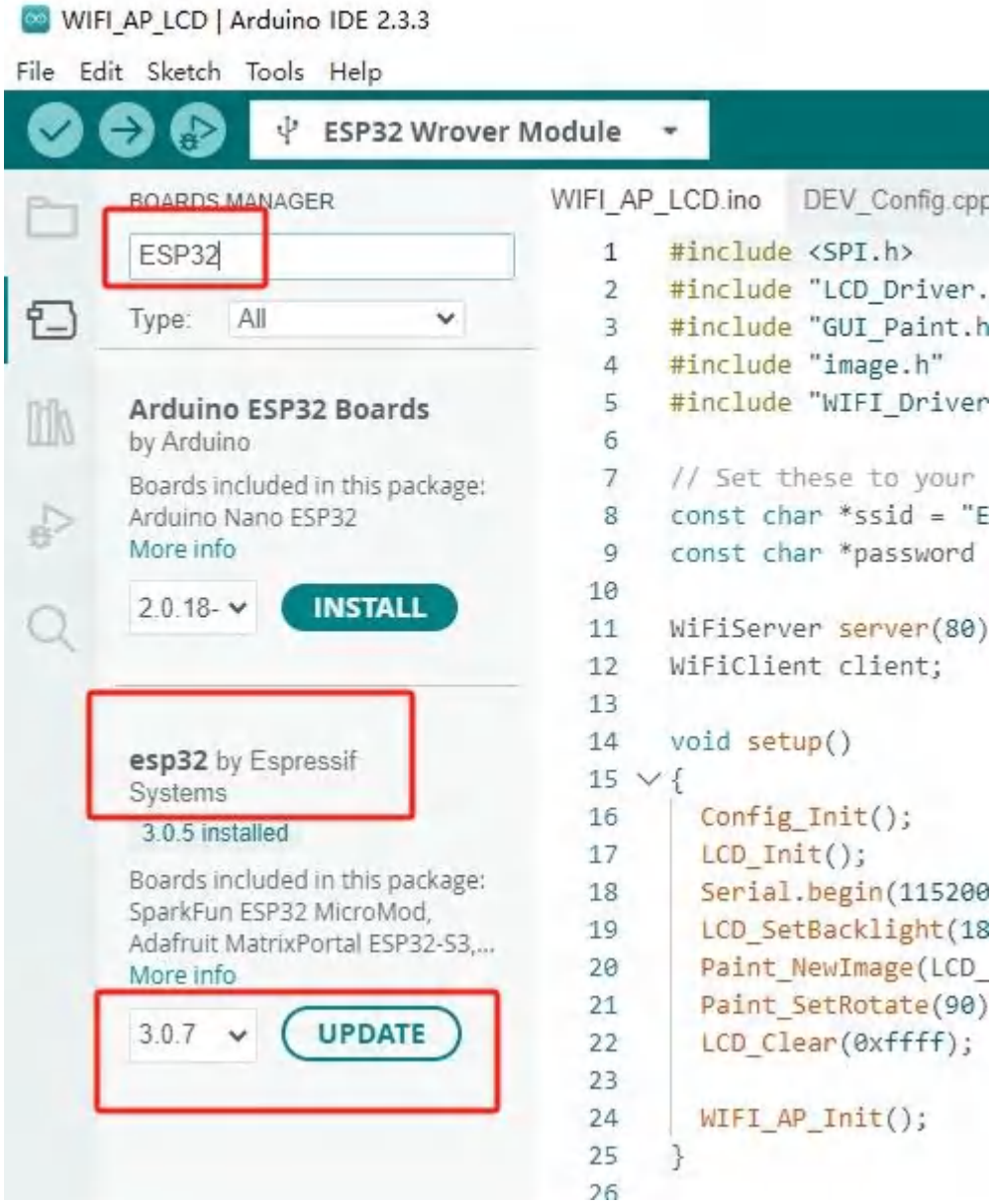
http://8.217.75.21/Industrial/Multilingual/AM03-FBBA0047-002-Product Library.zip



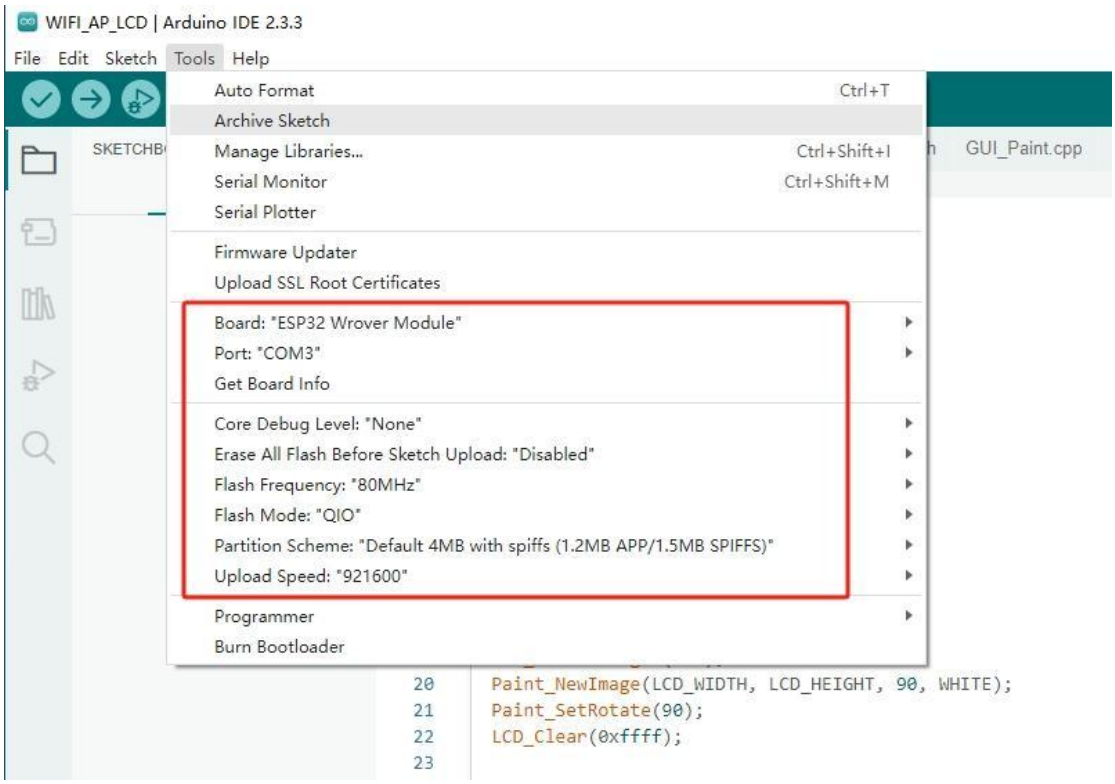
2. Add ESP32 development environment



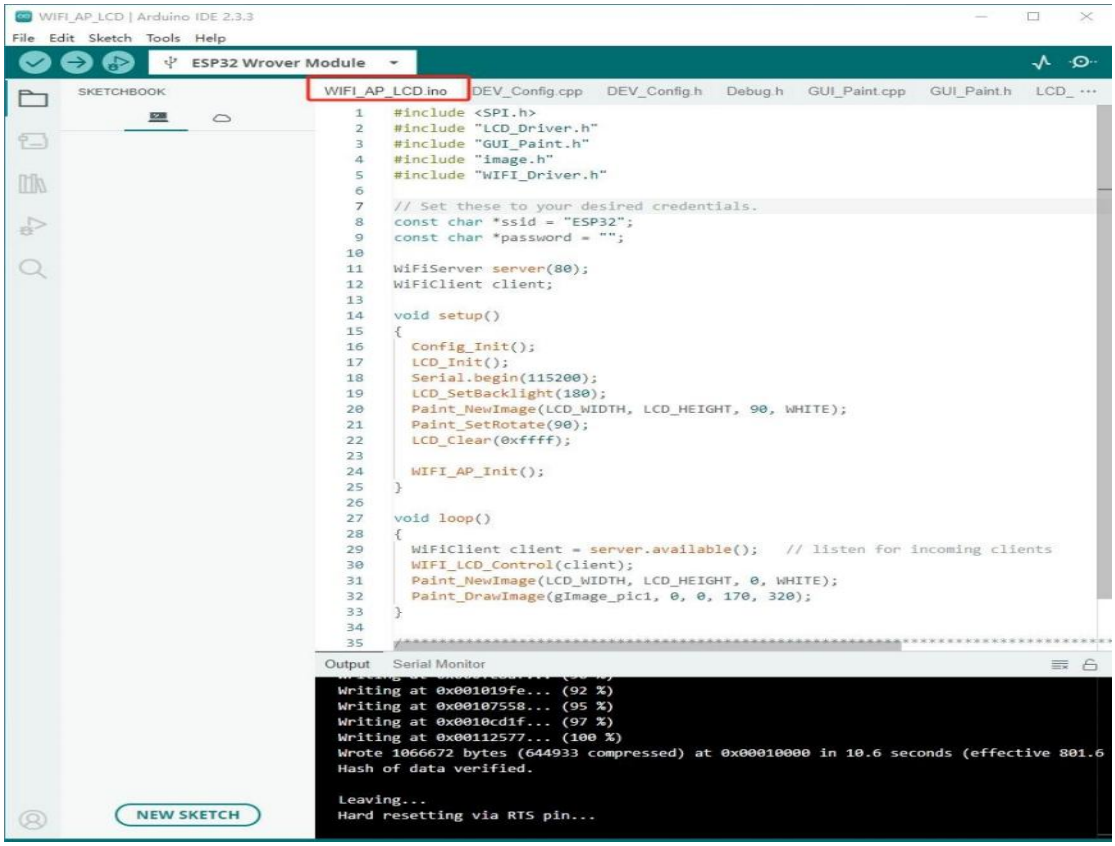
3. In the development board manager window, search for ESP32, install the development environment and add it.



4. After restarting the Arduino IDE software, configure the corresponding parameters of the development board, as shown in the red box below



5. Open the "WIFI_AP_LCD" routine and upload it to the development board



6. If the message "...._ ..." appears during downloading and cannot be burned, press and hold BOOT, re-plug the USB cable, and then release it, and then download the program again.

7. After the download is complete, HW-656 Server started 192.168.4.1 and acolor image will be displayed.

8. After connecting to the WiFi of ESP32, you can access this address to set the screen to display monochrome or images.